

Leo Xu

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TECHNICAL SKILLS

Languages: Java, Python, R, C/C++, C#, Kotlin, JavaScript, SQL, Bash, TypeScript, HTML/CSS

Data/ML: NumPy, Pandas, Matplotlib, scikit-learn, TensorFlow, PyTorch, Apache Spark

Frameworks: React, Node.js/Express, Qt5, Unity, JUnit, Mockito

Tools & Technologies: Git, Docker, RStudio, VS Code/ Visual Studio, Ansible, MongoDB, MySQL, Linux, Postman

EDUCATION

University of Waterloo

Waterloo, ON

Master of Data Science and Artificial Intelligence

Sept. 2025 – Present

Carleton University

Ottawa, ON

Bachelor of Computer Science Honours

Sept. 2019 – Apr. 2024

- Relevant Courses: Machine Learning, Optimization, Databases, Concurrent Systems, Graph Theory

EXPERIENCE

Android Developer

Jan. 2023 – Aug. 2023

General Motors

Markham, ON

- Proactively modernized an unmaintained Android intent-testing tool by replacing manual compilation workflows, cutting emulator iteration time by ~50% and significantly improving developer feedback loops
- Identified slow emulator feedback as a team-level bottleneck by comparing iteration latency across teams
- Improved consistency of intent-validation results across emulator runs, reducing false conclusions
- Implemented intent-driven UI state rendering for fast visual verification of emulator behavior
- Onboarded and guided new team members by documenting tooling and walking through debugging workflows

Data Scientist

May 2022 – Aug. 2022

Public Services and Procurement Canada

Ottawa, ON

- Migrated 100k+ procurement notices with a Python scraper (BeautifulSoup) replacing manual transfer
- Used Pandas to clean and validate procurement notice data by checking schema consistency, missing fields, and record counts, ensuring all public records were transferred correctly during the CanadaBuys migration
- Identified and documented 100+ defects during migration QA, enabling faster diagnosis by engineering teams
- Collaborated with developers and QA to prioritize high-impact issues and support migration readiness
- Supported the CanadaBuys platform launch through migration validation and post-release issue investigation

DevOps & Automation Engineer

Sept. 2021 – Dec. 2021

Royal Bank of Canada

Toronto, ON

- Built an ML-based anomaly detection workflow to monitor production API traffic, surfacing 50+ previously undetected usage and traffic anomalies during routine monitoring
- Routed anomaly signals to application support teams for faster investigation
- Tuned detection logic to reduce false positives and maintain alert usefulness for on-call engineers
- Automated alerting pipelines using Ansible to streamline incident response
- Extended monitoring coverage to legacy IMS databases on IBM z/OS, improving backend system visibility

PROJECTS

Used Car Price Predictor | *Python, Pandas, NumPy, scikit-learn, TensorFlow*

Sept. 2025 – Dec. 2025

- Built an end-to-end ML pipeline to predict used-car prices from structured vehicle and equipment data
- Trained and compared OLS, Ridge, Random Forest, XGBoost, and Wide & Deep models for tabular regression
- Achieved best performance with XGBoost, delivering lowest RMSE/MAE and highest R^2 on test data
- Applied 5-fold cross-validation to evaluate robustness and generalization across vehicle brands
- Analyzed feature importance and price lift to quantify brand and equipment effects on valuation

Social Dynamics in Survival Scenarios | *Python*

Jan. 2024 – April. 2024

- Leveraged OpenAI Whisper + Pyannote-Audio to generate accurate speaker-labeled transcripts
- Processed dialogue into interaction datasets, categorizing relationship valence and mapping evolving social ties
- Engineered NLP pipeline in Python to quantify sentiment and model interactions as weighted graphs
- Built dynamic social graphs using Python, NetworkX, and Matplotlib to analyze group cohesion and leadership
- Applied graph theory metrics to model and predict team performance